

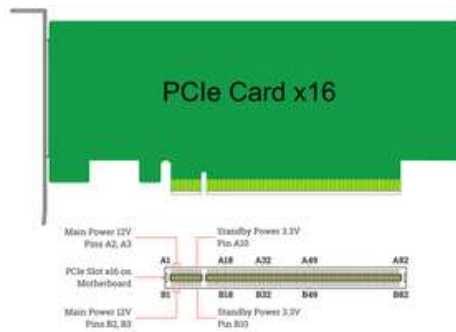
Transfer podataka

IO kod akceleratora

Transfer podataka

- Deo za obradu podataka čeka da mu pristignu podaci
- Svaki „handshake“ protokol
 - Smanjuje propusni opseg (bandwidth)
 - Povećava kašnjenje (latency)
- Mogući izvori podataka na FPGA-u akceleratoru
 - PCIe magistrala
 - Mreža
 - DDR ili HBM memorija

Bandwith

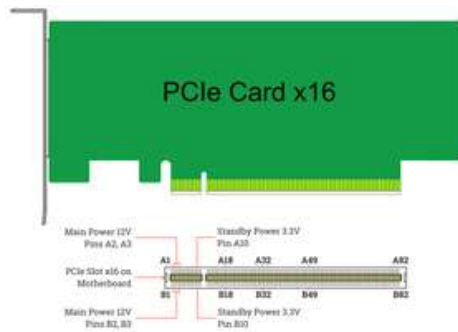


Verzija	Bandwith
3.0 x16	15.75 GB/s
4.0 x16	31.5 GB/s
5.0 x16	63 GB/s

Verzija	Bandwith
DDR4-3200	25600 MB/s
DDR5-6400	31.5 GB/s

Verzija	Bandwith
25G-1	25 Gb/s
100G-4	100 Gb/s
400G-8	400 Gb/s

Latency



~ 1us

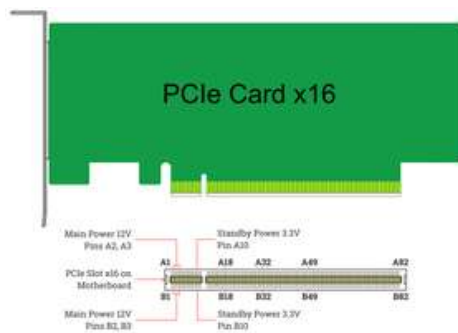


~ 10 ns



~ 1us

Širina reči



512 bit



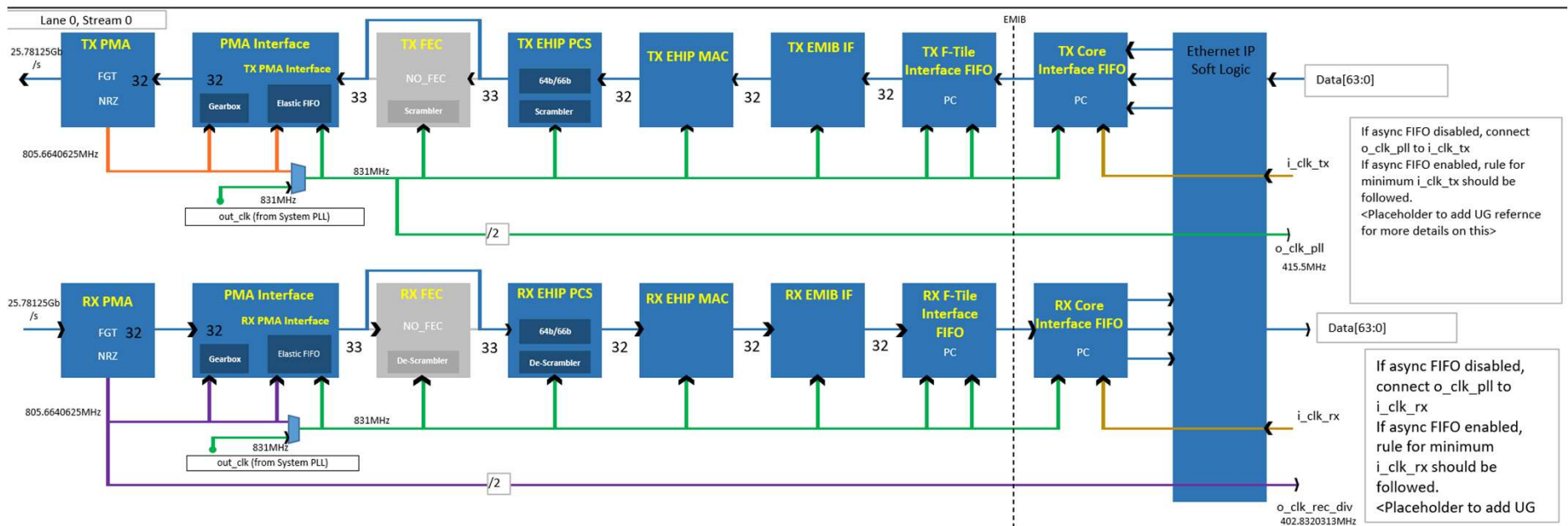
64 bit



64, 512, 1024 bit

Tehnologija IO pinova

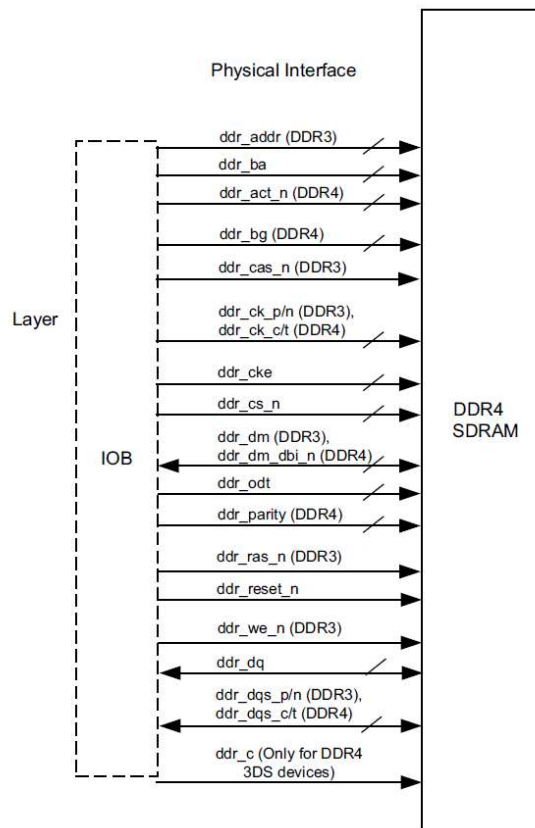
- PCIe, mreža:
- SERDES



The picture is Intel property taken from F-Tile Architecture and PMA and FEC Direct PHY IP User Guide

Tehnologija IO pinova

- DDR:
 - Paralelni interfejs

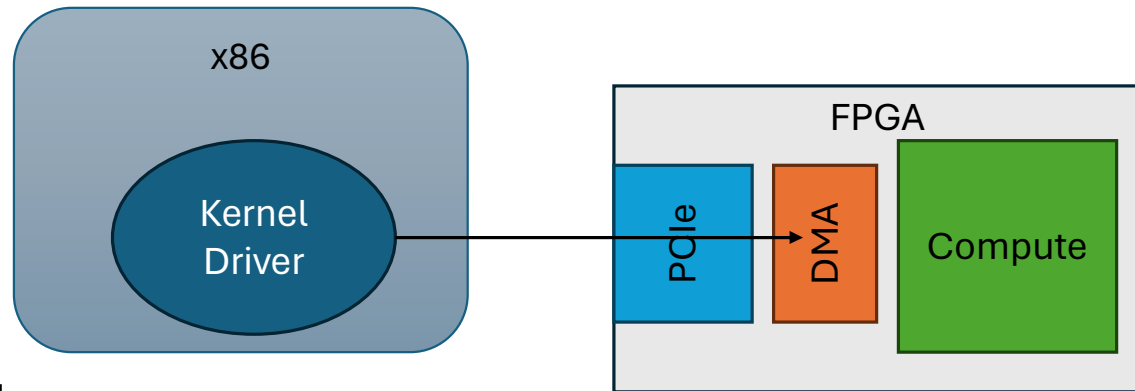


The picture is AMD property taken from UltraScale Architecture-Based FPGAs Memory

Implementacija kontrolera

- Već ugrađeni u FPGA (Hard IP) – nije ih potrebno realizovati
- Obično pružaju standardizovan interfejs
 - AXI4
 - Avalon
 - Segmentirani interfejs za šire reči
- Pristojna podrška za simulaciju i testiranje
- Pristojno dokumentovano

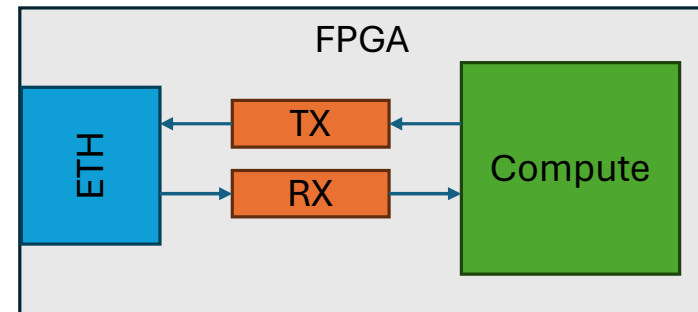
PCIe



- Paketski transfer
- Read i write komande
- Reč je 512 bit, a x86 instrukcija može maks 256 bita transfer
 - DMA kontroler je obavezan (može i sa IOMMU)
 - DMA zahteva drajver za kernel
- Legacy zahtev za boot od 120ms
 - Prvo se učitava na FPGA mali dizajn
 - Tek kada se sistem podigne veliki dizajn akceleratora
 - Tokom razvoja greška pri PCIe transferu zahteva restart sistema

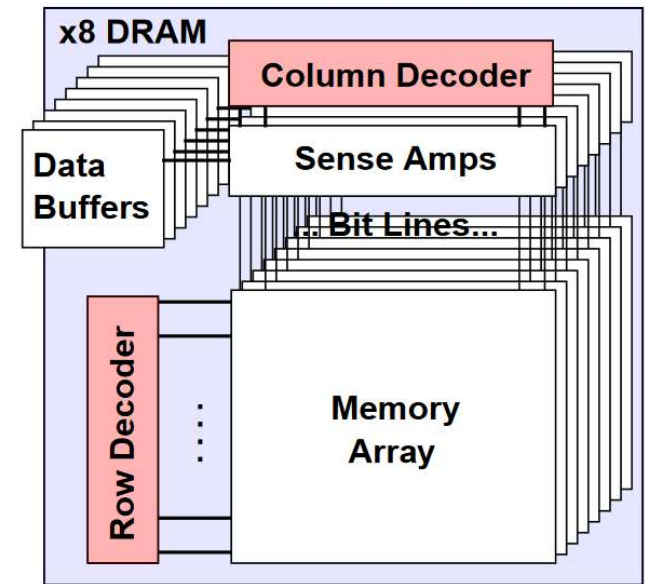
Mreža

- Paketski transfer
- RX i TX kanali (istovremeni transfer)
- Limiti => sve se radi na clock od ETH ~400MHz
 - RX nema mogućnost za pauziranje
 - TX paket mora bez pauze da se šalje
- TCP protokol
 - Ima „handshake“ sa ponovnim slanjem
 - Na osnovu broja sekvence mogu paketi da se premeštaju
 - Checksum je na početku paketa
- Na sreću osim fizičkog paket ne mora da ima ništa od viših slojeva



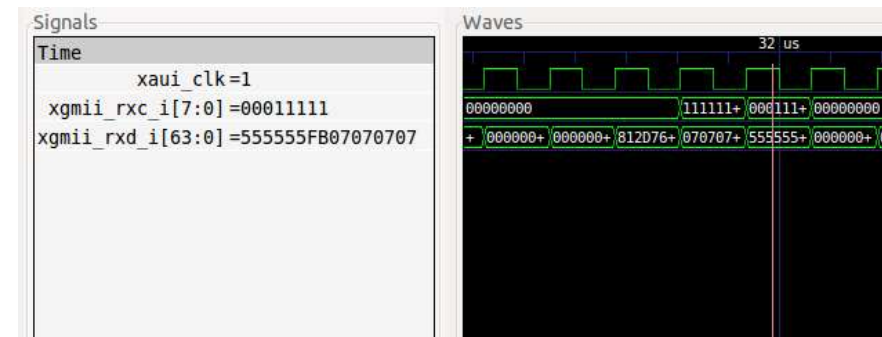
DDR

- Transfer jedne reči
- Manja latencija u odnosu na ostale
 - Začkoljica – ima refresh na 7,8us
- Veliki prostor (i preko 64GB po akcelertoru)
- Maksimalna brzina je za sekvencijalni pristup
- Upis je sporiji od čitanja



Koji koristiti?

- Višestruka čitanje i upis
 - DDR
- Potreba da se nešto odradi uz pomoć x86
 - PCIe
- Chip to Chip transfer
 - Mrežni interfejs
 - Najbolje prilagođeni protokol bez stanja
 - Može i bez MAC sloja => XGMII interface
- Podaci sa interneta
 - Mrežni interfejs

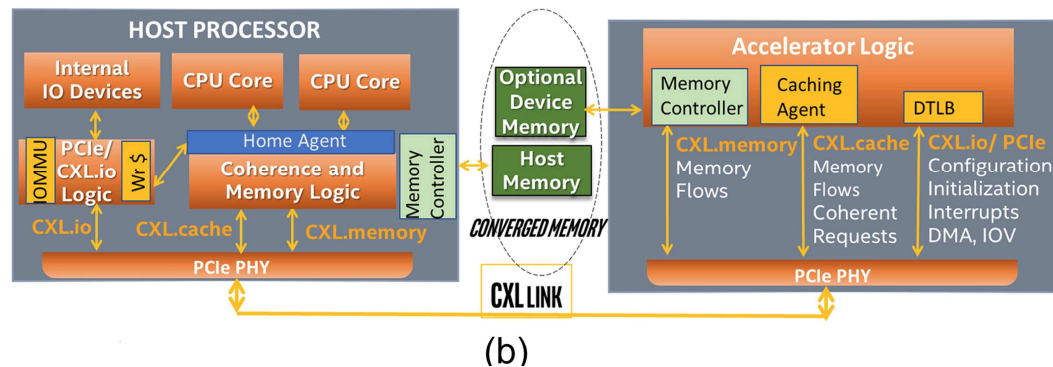
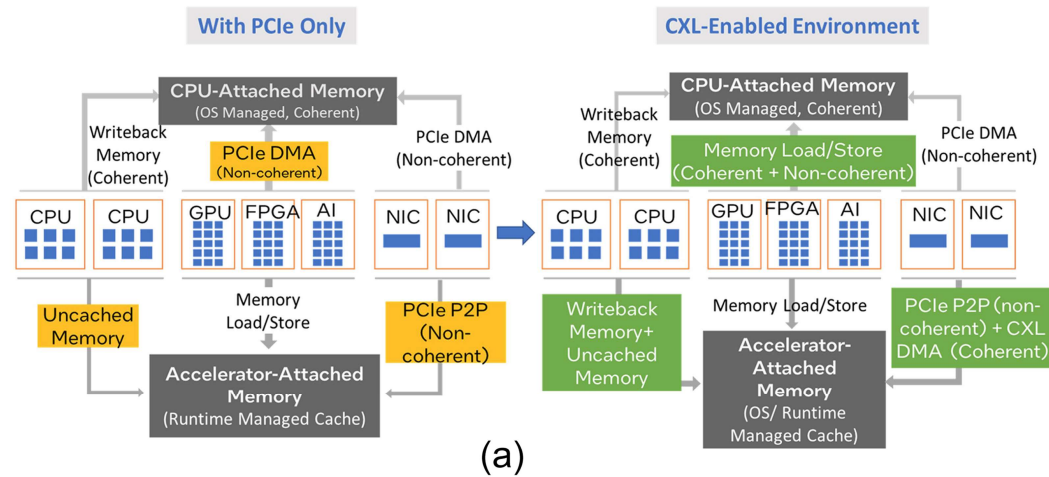


Primer dostupnih resursa

Product Line		AGI 019	AGI 023	AGI 022	AGI 027	AGI 035	AGI 040	AGI 041
Resources	Logic elements (LEs)	1,918,975	2,308,080	2,208,075	2,692,760	3,540,000	4,047,400	4,000,672
	Adaptive logic modules (ALMs)	650,500	782,400	748,500	912,800	1,200,000	1,372,000	1,356,160
	ALM registers	2,602,000	3,129,600	2,994,000	3,651,200	4,800,000	5,488,000	5,424,640
	High-performance crypto blocks	2	2	0	0	4	4	4
	eSRAM memory blocks	1	1	0	0	3	3	2
	eSRAM memory size (Mb)	18	18	0	0	54	54	36
	M20K memory blocks	8,500	10,464	10,900	13,272	14,931	19,908	17,136
	M20K memory size (Mb)	166	204	212	259	292	389	335
	MLAB memory count	32,525	39,120	37,425	45,640	60,000	68,600	67,808
	MLAB memory size (Mb)	20	24	23	28	37	42	42
	Fabric PLL	5	5	12	12	6	6	8
	I/O PLL	10	10	16	16	12	12	16
	Variable-precision digital signal processing (DSP) blocks	1,354	1,640	6,250	8,528	9,594	12,792	0
	18 x 19 multipliers	2,708	3,280	12,500	17,056	19,188	25,584	0
	Single-precision or half-precision tera floating point operations per second (TFLOPS)	2.4 / 4.9	2.4 / 4.9	9.4 / 18.8	12.8 / 25.6	14.3 / 28.7	19.1 / 38.3	0
Maximum Available Device Resources	Maximum EMIF x72 ¹	3	3	4	4	4	4	4
	Maximum differential (RX or TX) pairs	240	240	360	360	288	288	372
	Maximum AIB Interfaces	4	4	4	4	6	6	4
	Memory devices supported	DDR4 and QDR IV						
	Secure device manager (SDM)	Provides SHA-384 bitstream integrity, ECDSA 256/384 bitstream authentication, AES-256 bitstream encryption, physically unclonable function (PUF) protected key storage, side-channel attack resistance, SPDMM attestation, cryptographic services, physical anti-tamper support						
Tile Resources	Hard processor system	Quad-core 64 bit Arm Cortex®-A53 up to 1.50 GHz with 32 KB I/D cache , Neon® coprocessor, 1 MB L2 cache, direct memory access (DMA), system memory management unit, cache coherency unit, hard memory controllers, USB 2.0x2, 1G EMAC x3, UART x2, serial peripheral interface (SPI) x4, I2C x5, general purpose timers x7, watchdog timer x4				n/a		HPS support ²
	F-Tile	PCI Express® (PCIe®) hard IP block (4.0 x16) or Bifurcateable 2x PCIe 4.0 x8 (EP) or 4x 4.0 x4 (RP) Transceiver channel count : - 4 channels at 116 Gbps (PAM4) / 58 Gbps (NRZ) - 16 channels at 32 Gbps (NRZ) /12 channels at 58 Gbps (PAM4) - RS & KP FEC Advanced networking support: - Bifurcatable 400 GbE hard IP block (10/25/50/100/200/400 GbE FEC/PCS/MAC) - Bifurcatable 200 Gb hard IP block (10/25/50/100/200 Gbs FEC/PCS) IEEE 1588 support PMA direct						
	R-Tile	Compute Express Interface (CXL) - Link width x16 lanes, x8 lanes PCIe hard IP block (5.0 x16) or Bifurcateable 2x PCIe 5.0 x8 (EP) or 4x 5.0 x4 (RP) Virtualization (SR-IOV) supporting 8 PFs/2k VFs Scalable IOV VirtIO support Precise time management PIPE direct						

The picture is Intel property taken from Agilex™ 7 FPGAs and SoCs I-Series Product Table

Industrijska rešenja - CXL



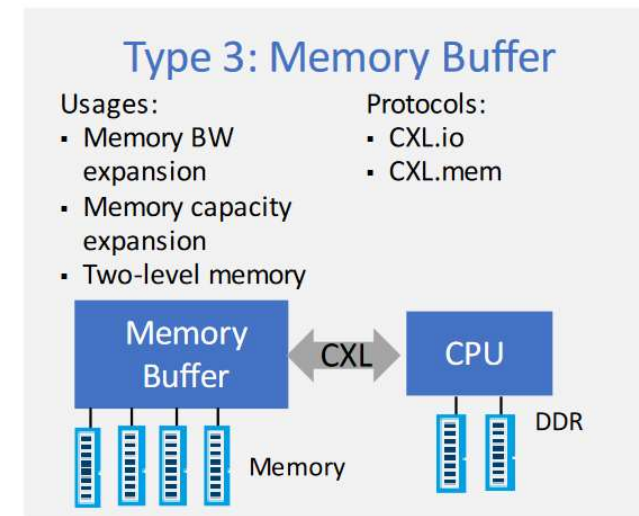
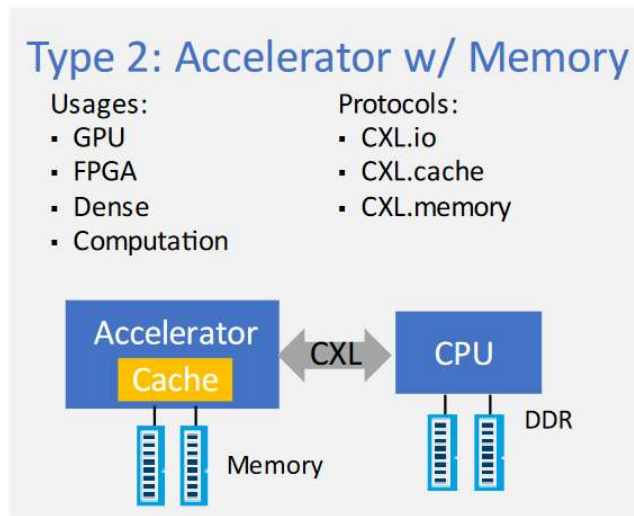
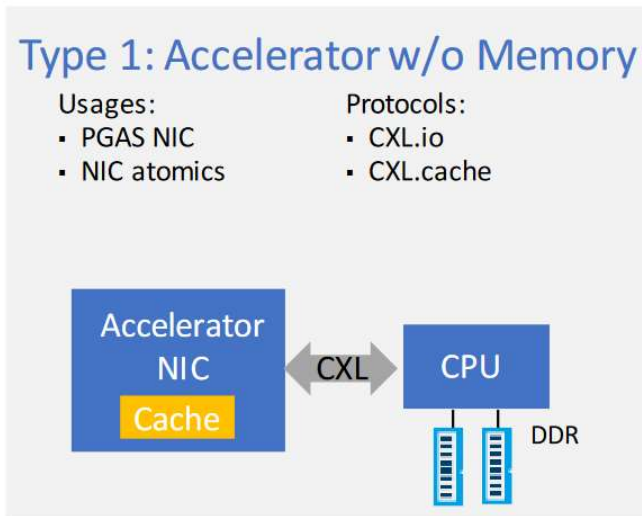
The picture is from [1]

CXL razlozi za uvođenje [3]

1. Uređaj i CPU imaju nekoherentan pristup memoriji
 - CPU kešira podatke iz svoje memorije, ali ne i iz memorije uređaja
 - Uređaj nema svest o postojanju kopije podatka u keš memoriji CPU-a
2. Skaliranje memorije
 - Limit pinova
 - Cena po bitu u poslednje vreme sporo opada
 - PCIe žice mogu da se produže (retimers)
3. Neefikasnost zbog loše preraspodele resursa
 - Server ne može da pristupi uređajima koje je drugi zauzeo, a ne koristi ih
4. Transferi podataka na finom nivou granulacije su spori
 - 4 KB se prebacuje za 2us na linku od 400 Gb/s, a latencija je 10us

CXL 1.x specifikacija

- Rešava probleme 1 i 2 uz pomoć tri različite konfiguracije
- Tri protokola CXL.io, CXL.mem i CXL.cache



The picture is from [3]

CXL protokoli

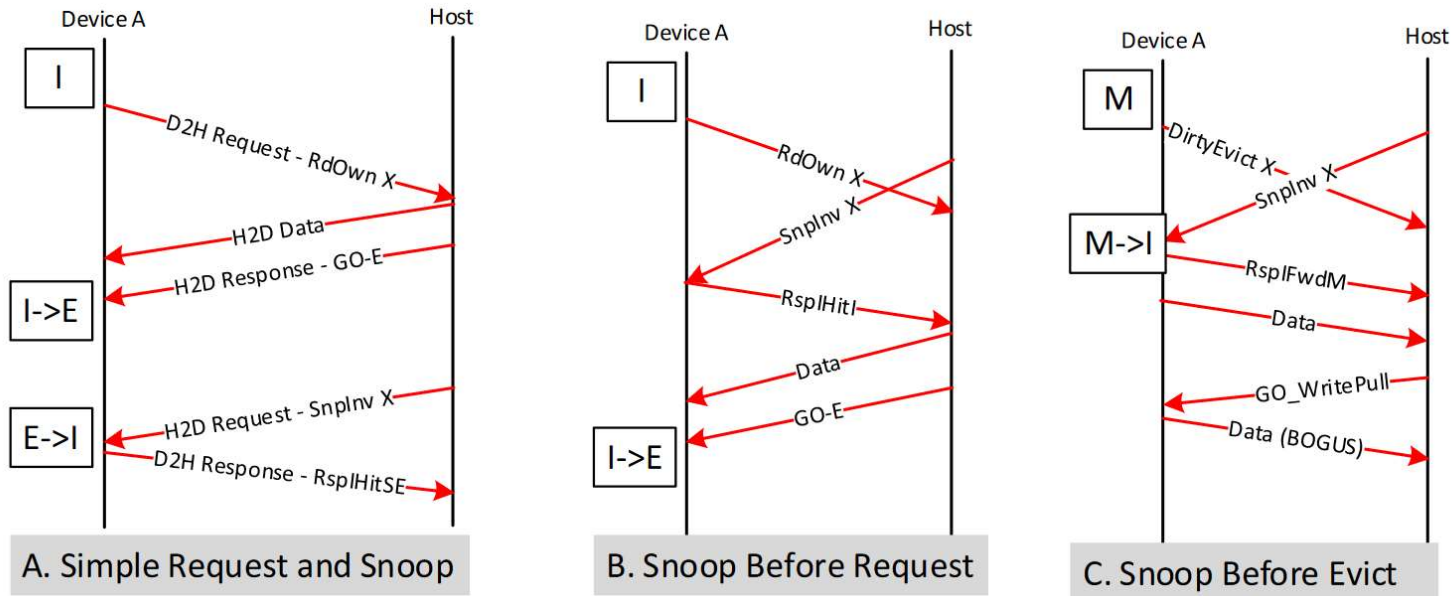
- CXL.io
 - U potpunosti koristi PCIe stek
 - Služi za: konfiguraciju, DMA transfere i nekoheretne upise/čitanja
- CXL.cache i CXL.mem
 - Koriste samo fizički sloj PCIe steka, ostalo implementiraju sami
 - Osnovna jedinica transfera Flit (68B)

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Byte 0	PROTID (2B)		Flit Hdr (4B)				Header Slot (12B)									
Byte 16					G-Slot (16B)											
Byte 32					G-Slot (16B)											
Byte 48					G-Slot (16B)											
Byte 64	Slot 4 - 16B (G-Slot)				CRC(2B)		PROTID (2B)		New All Data Flit Starts here							

The picture is from [2]

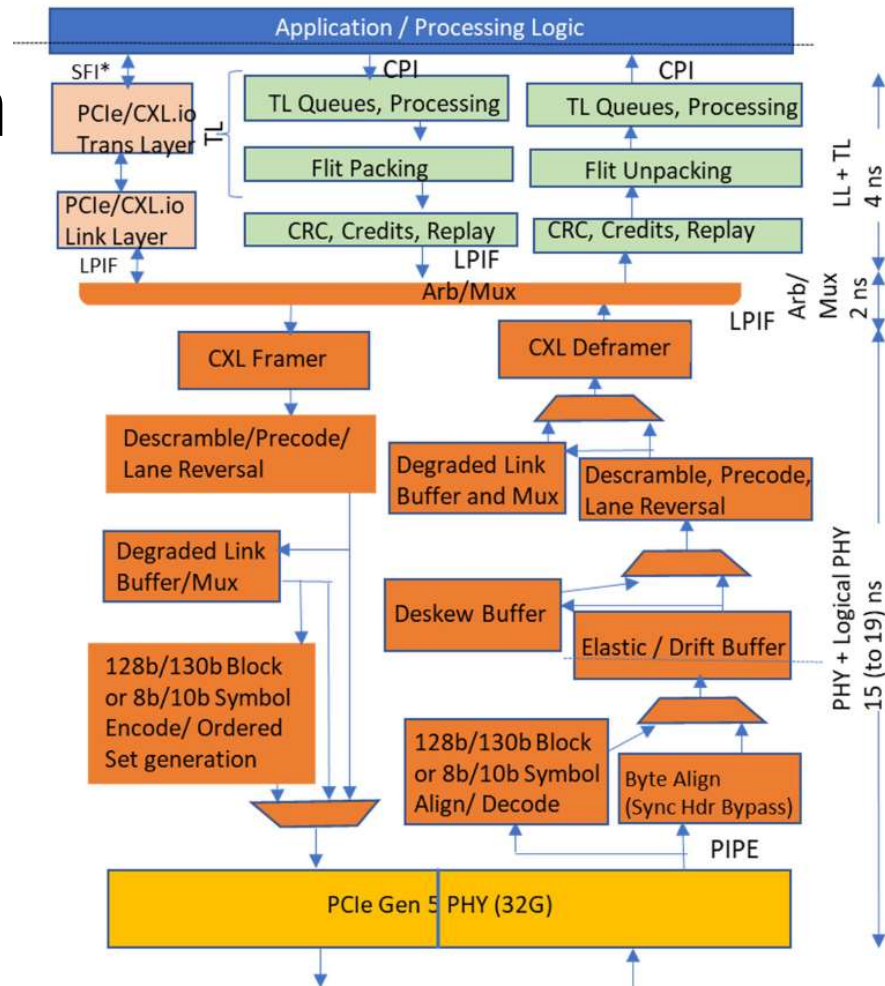
CXL Koherencija

- Koristi se MESI protokol sa keš linijama od 64B
- Komunikacija je samo između CPU i uređaja



The picture is from [3]

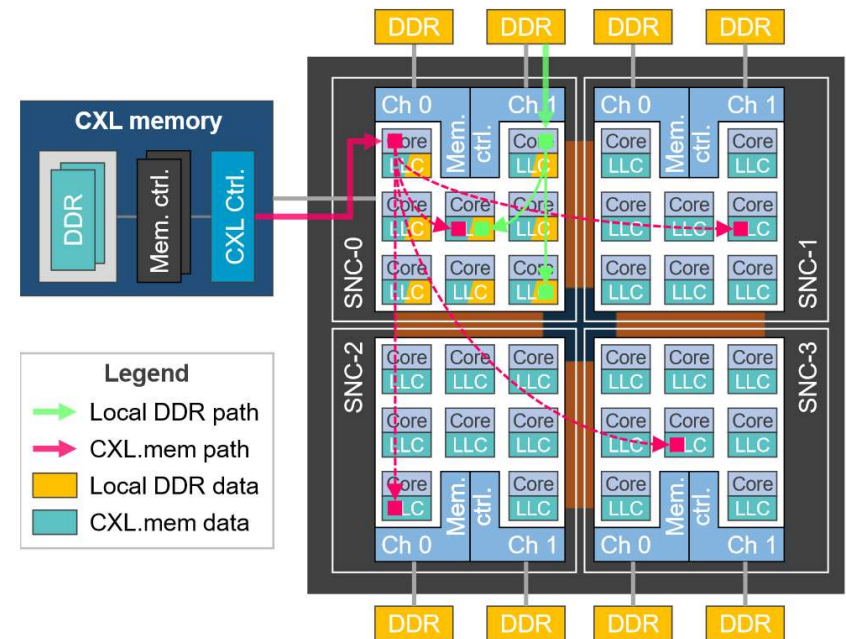
CXL Mikroarhitektura



The picture is from [2]

CXL testiranje na komercijalnim uređajima [1]

- Testirano proširenje memorije
 - Bolje nego NUMA
- Pogodno za neke vrste aplikacija
 - Zahtevaju veliku propustnost memorije
- Veća fleksibilnost keširanja
 - Podaci sa uređaja mogu u bilo koji keš
 - CPU podaci samo u neke keš memorije

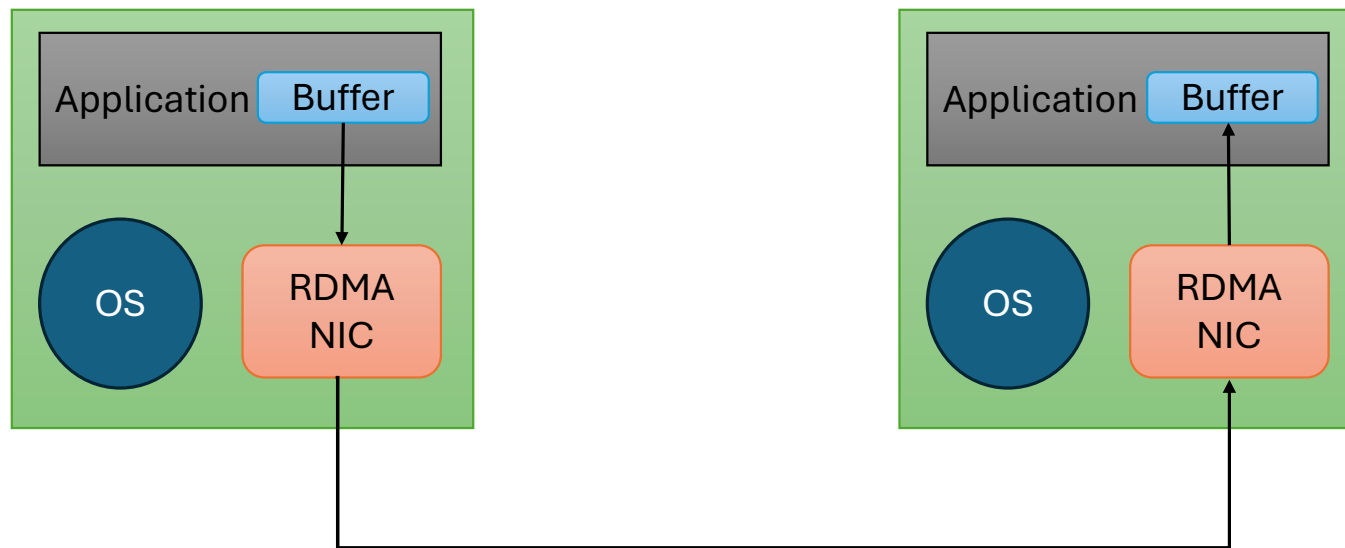


The picture is from [1]

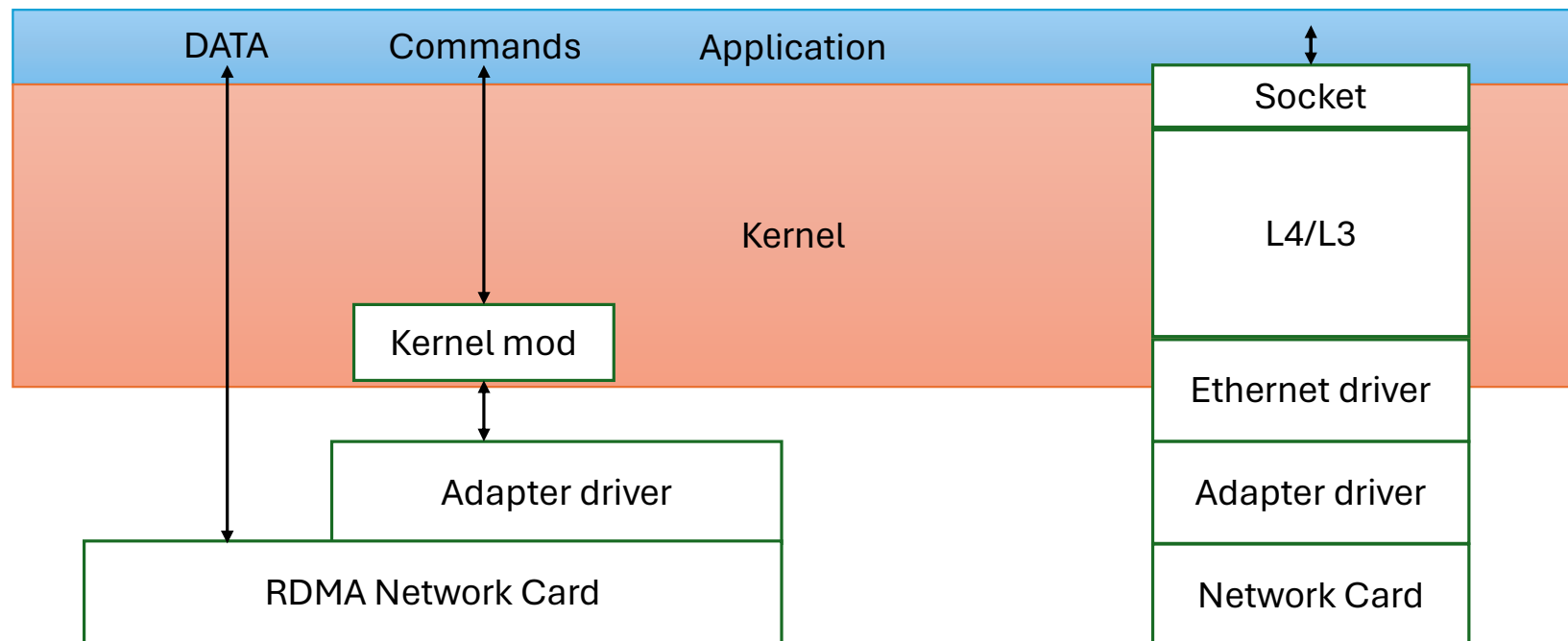
Industrijska rešenja - RDMA

- Komunikacija CPU-a i udaljenog FPGA-a preko mreže
- Prednosti u odnosu na klasičan stek:
 - Nema kopiranja podataka
 - Zaobilazi se kernel
 - CPU skoro da ne učestvuje u transferu
 - Niska latencija za kratke poruke
 - Visoka propusnost
- Inicijalno napravljen za Infiniband, sad postoji i za Ethernet

RDMA mehanizam



RDMA poređenje sa klasičnim stekom



RDMA implementacija

- Najčešće se koristi UDP kao noseći protokol
- Može imati podršku za pouzdan transfer
- Sav posao radi hardver mrežne kartice
- Asinhrona komunikacija

Reference

- [1] D. D. Sharma, "Compute Express Link (CXL): Enabling Heterogeneous Data-Centric Computing With Heterogeneous Memory Hierarchy," in *IEEE Micro*, March-April 2023
- [2] Y. Sun et al, " Demystifying CXL Memory with Genuine CXL-Ready Systems and Devices," MICRO 2023
- [3] D. D. Sharma, et al. "An introduction to the compute express link (cxl) interconnect," *arXiv* 2023.